

RIDA MELKAOUI

INDUSTRIAL ENGINEER | DATA, AI & BUSINESS INTELLIGENCE

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LinkedIn | GitHub | Portfolio

PROFILE

Industrial & Production Engineer who combines supplier-quality ownership with data analytics, BI and applied AI. At Magna, leads electronics and metal supplier quality across international programs and builds the operating views used to track launch readiness, issues and actions. Translates industrial problems into measurable decision products with Python, SQL and Power BI.

PROFESSIONAL EXPERIENCE

MAGNA INTERNATIONAL Supplier Quality Engineer — Electronics & Metal Commodities - Own supplier-quality performance for electronics and metal commodities across China, South Korea, Spain, Slovakia, Germany and Morocco, using launch readiness, 8D/CAPA status and open-issue risk to focus escalation. - Developed X-Rey, a full SQE BI and workflow system covering nomination, Safe Launch and SOP; unified ownership, milestones, PPAP/DVP evidence, critical parts and open issues across 6 control modules and 4 lifecycle gates. - Implemented KPI views and an automated notification pipeline for missing baselines, overdue actions and gate risks, keeping responsibilities and deadlines visible across the supplier portfolio.	Kenitra, Morocco Mar 2026 — Present
STELLANTIS R&D Engineering Intern — Data Analytics, Process Optimization & AI Mapped the E-BOM/M-BOM anomaly process and built a Python application plus structured Databricks model for detection, prioritization and traceability; reduced the validated treatment cycle from 2 days to under 3 minutes (–98% manual handling, 100% project test-case reliability).	Casablanca, Morocco Feb 2025 — Aug 2025
SANY — FORGE DE BAZAS Engineering Intern — Process Digitalization & Maintenance Built a Unity/NLP assistant that structured access to maintenance and spare-parts knowledge, replacing fragmented shop-floor search with a guided information flow.	Casablanca, Morocco Jul 2024 — Sep 2024
3D SMART FACTORY Data Analyst Intern Structured Power BI and Excel reporting around OEE, FPY, scrap and downtime, giving Lean reviews a single view of production loss drivers and bottlenecks.	Mohammedia, Morocco Aug 2023 — Sep 2023

PROJECTS

Demand / Order — Forecasting & replenishment Built a cutoff-safe Python forecasting and replenishment engine to decide what to stock and where to review exceptions. It modeled 900 M5 item-store series, benchmarked candidates against calibrated seasonal baselines and converted the best signal into inventory scenarios—improving WAPE by 4.5% and simulating 1,528 fewer lost units at 96.7% fulfilled demand on a 28-day holdout.
Route / Control — Optimization & operations Built a dispatch control tower that learned zone-order preferences from 6,112 public histories, combined directed travel times with constrained local search and surfaced exceptions. On an unseen 13-route cohort, it cut simulated drive time by 2.8%, preserved 100% timed-package adherence and eliminated 332 zone re-entries.
Agent / Proof — AI evaluation & observability Built an AI-agent evaluation lab over 3,336 official tool-use trajectories, measuring pass@k, repeatability, cost, latency and failure modes before release. Its pinned GPT-4.1 telecom cohort revealed a 14.9-point gap between one-shot and repeatable success.

EDUCATION

Mohammed VI International Civil Aviation Academy (AIAC) Engineering Degree Industrial & Production Engineering	Nouaceur, Morocco 2022 — 2025
Salmane El Farissi Preparatory Classes Preparatory Studies MPSI - MP	Salé, Morocco 2020 — 2022

SKILLS

. DATA & ANALYTICS	Python · SQL · Pandas · NumPy · scikit-learn · statistics · forecasting · experimentation · data modeling · Databricks
. BI & DELIVERY	Power BI · DAX · Excel · Power Query · Tableau · React · TypeScript · FastAPI · Git · Docker · KPI systems · automation
. INDUSTRIAL & QUALITY	Supplier/process performance · PPM · OEE · FPY · 8D/CAPA · FMEA · root-cause analysis · Lean · launch/SOP governance

LANGUAGES

Arabic Native	French Fluent	English C1 TOEIC 855 EF SET 62/100
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CERTIFICATIONS

IBM Data Analyst Professional Certificate IBM / Coursera 2026	EF SET English Certificate C1 EF SET 62/100	TOEIC Listening & Reading C1 Score 855
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